```
# Task1
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

food = ["Rice", "Milk", "Egg"]
calories = [200, 150, 100]

df = pd.DataFrame({"Food": food, "Calories": calories})

total = np.sum(df["Calories"])
print("Total Calories:", total)

plt.pie(df["Calories"], labels=df["Food"], autopct="%1.1f%%")
plt.title("Calorie Intake")
plt.show()

# Task2
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

products = ["Pen", "Book", "Bag"]
quantity = [50, 30, 20]

df = pd.DataFrame({"Product": products, "Quantity": quantity})

total = np.sum(df["Quantity"])
print("Total Stock:", total)

plt.bar(df["Product"], df["Quantity"])
plt.title("Stock Levels")
plt.show()
```

```
# Task3
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

players = ["A", "B", "C"]
scores = [80, 60, 90]

df = pd.DataFrame({"Player": players, "Score": scores})

print("Highest Score:", np.max(df["Score"]))

plt.bar(df["Player"], df["Score"])
plt.title("Game Scores")
plt.show()

# Task4
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

hours = ["9AM", "10AM", "11AM", "12PM"]
vehicles = [50, 80, 120, 90]

df = pd.DataFrame({"Time": hours, "Vehicles": vehicles})

peak = df.iloc[np.argmax(df["Vehicles"])]
print("Peak Time:", peak["Time"])

plt.plot(df["Time"], df["Vehicles"])
plt.title("Traffic Analysis")
plt.show()
```

Q5.

TP = 50, FP = 5, FN = 10, TN = 35

Total = 50 + 10 + 5 + 35 = 100

Accuracy = (50 + 35) / 100 = 0.85

Precision = 50 / (50 + 5) = 0.909

Recall = 50 / (50 + 10) = 0.8333

F1 Score = 2 × (0.909 × 0.8333) / (0.909 + 0.8333)

F1 Score = 2 × 0.7574 / 1.7423

F1 Score = 0.8694

Q6.

Total = 40 + 5 + 5 + 10 + 30 + 10 + 5 + 5 + 40 = 150

Accuracy = (40 + 30 + 40) / 150 = 0.7333

Precision (A) = 40 / (40 + 10 + 5) = 0.7272

Recall (A) = 40 / (40 + 5 + 5) = 0.8

F1 Score (A) = 2 × (0.7272 × 0.8) / (0.7272 + 0.8)

F1 Score (A) = 2 × 0.5817 / 1.5272

F1 Score (A) = 0.7618

Precision (B) = 30 / (5 + 30 + 5) = 0.75

Recall (B) = 30 / (10 + 30 + 10) = 0.6

F1 Score (B) = 2 × (0.75 × 0.6) / (0.75 + 0.6)

F1 Score (B) = 2 × 0.45 / 1.35

F1 Score (B) = 0.6666

Precision (C) = 40 / (5 + 10 + 40) = 0.7272

Recall (C) = 40 / (5 + 5 + 40) = 0.8

F1 Score (C) = 2 × (0.7272 × 0.8) / (0.7272 + 0.8)

F1 Score (C) = 2 × 0.5817 / 1.5272

F1 Score (C) = 0.7618

Q7.

TP = 20, FP = 10, FN = 80, TN = 890

Total = 20 + 80 + 10 + 890 = 1000

Accuracy = (20 + 890) / 1000 = 0.91

Precision = 20 / (20 + 10) = 0.6666

Recall = 20 / (20 + 80) = 0.2

F1 Score = 2 × (0.6666 × 0.2) / (0.6666 + 0.2)

F1 Score = 2 × 0.1333 / 0.8666

F1 Score = 0.3076

Q8.

TP = 45, FP = 15, FN = 5, TN = 35

Total = 45 + 5 + 15 + 35 = 100

Accuracy = (45 + 35) / 100 = 0.8

Precision = 45 / (45 + 15) = 0.75

Recall = 45 / (45 + 5) = 0.9

F1 Score = 2 × (0.75 × 0.9) / (0.75 + 0.9)

F1 Score = 2 × 0.675 / 1.65

F1 Score = 0.8181

```
# NUMPY TASKS
import numpy as np

# Task 1
arr = np.arange(1,11)
print(arr)

# Task 2
data = np.array([10,20,30,40,50])
print("Mean:", np.mean(data))
print("Median:", np.median(data))
print("Std Dev:", np.std(data))

# Task 3
rand_nums = np.random.rand(10)
print(rand_nums)

# Task 4
a = np.arange(1,7)
reshaped = a.reshape(2,3)
print(reshaped)

# Task 5
print("Max:", np.max(data))
print("Min:", np.min(data))

# Task 6
zeros = np.zeros((3,3))
ones = np.ones((3,3))
print(zeros)
print(ones)
```

```
# PANDAS TASKS
import pandas as pd

# Task 1
df = pd.DataFrame({
    "Name":["Asha","Ravi","John","Meena"],
    "Marks":[85,90,78,88]
})
print(df)

# Task 2
df["Grade"] = ["A","A+","B","A"]
print(df)

# Task 3
sorted_df = df.sort_values(by="Marks")
print(sorted_df)

# Task 4
data = pd.read_csv("employee_data.csv")
print(data.head(25))

# Task 5
clean = data.dropna()
print(clean)

# Task 6
data.fillna(data.median(numeric_only=True), inplace=True)
print(data)

# MATPLOTLIB TASKS
import matplotlib.pyplot as plt

# Task 1
months = ["Jan","Feb","Mar","Apr","May"]
```

```
sales = [100,150,200,180,220]
plt.plot(months,sales)
plt.title("Monthly Sales")
plt.show()

# Task 2
students = ["A","B","C","D"]
marks = [80,90,70,85]
plt.bar(students,marks)
plt.title("Student Marks")
plt.show()

# Task 3
height = [150,160,170,180]
weight = [50,60,65,75]
plt.scatter(height,weight)
plt.title("Height vs Weight")
plt.show()

# Task 4
ages = [18,20,22,22,23,25,27,30]
plt.hist(ages)
plt.title("Age Distribution")
plt.show()

# Task 5
plt.plot(months,sales,label="Sales")
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.legend()
plt.show()

# Task 6
sales2 = [90,120,160,170,200]
```

```
plt.plot(months,sales,label="2024")
plt.plot(months,sales2,label="2025")
plt.legend()
plt.show()

# Task 7
plt.plot(months,sales,color="red",marker="o")
plt.grid(True)
plt.show()
```

 # TKINTER TASKS

```
import tkinter as tk

# Task 1
root = tk.Tk()
root.title("Basic Window")
root.geometry("300x200")
root.mainloop()

# Task 2
root = tk.Tk()
label = tk.Label(root,text="Hello")
label.pack()
btn = tk.Button(root,text="Click")
btn.pack()
root.mainloop()

# Task 3
def calc(op):
    a=int(e1.get())
    b=int(e2.get())
    if op=="add": r=a+b
    if op=="sub": r=a-b
    if op=="mul": r=a*b
    if op=="div": r=a/b
```

```
    result.config(text=str(r))

root=tk.Tk()
e1=tk.Entry(root)
e2=tk.Entry(root)
e1.pack()
e2.pack()
tk.Button(root,text="Add",command=lambda:calc("add")).pack()
tk.Button(root,text="Sub",command=lambda:calc("sub")).pack()
tk.Button(root,text="Mul",command=lambda:calc("mul")).pack()
tk.Button(root,text="Div",command=lambda:calc("div")).pack()
result=tk.Label(root,text="")
result.pack()
root.mainloop()

# Task 4
root=tk.Tk()
tk.Label(root,text="Username").pack()
tk.Entry(root).pack()
tk.Label(root,text="Password").pack()
tk.Entry(root,show="*").pack()
tk.Button(root,text="Login").pack()
root.mainloop()

# Task 5
def msg():
    label.config(text="Button Clicked")

root=tk.Tk()
btn=tk.Button(root,text="Click",command=msg)
btn.pack()
label=tk.Label(root,text="")
label.pack()
root.mainloop()
```

```
# Task 6
root=tk.Tk()
txt=tk.Entry(root)
txt.pack()
root.mainloop()

# Task 7
root=tk.Tk()
text=tk.Text(root)
text.pack()
root.mainloop()

# Task 8
root=tk.Tk()
menu=tk.Menu(root)
root.config(menu=menu)
filemenu=tk.Menu(menu)
menu.add_cascade(label="File",menu=filemenu)
filemenu.add_command(label="New")
filemenu.add_command(label="Open")
root.mainloop()
```

 # PYQT TASKS

```
import sys
from PyQt5.QtWidgets import QApplication,QWidget,QLabel,QPushButton,QVBoxLayout,QLineEdit

app=QApplication(sys.argv)

# Task 1
win=QWidget()
win.setWindowTitle("Basic Window")
layout=QVBoxLayout()
layout.addWidget(QLabel("Hello PyQt"))
win.setLayout(layout)
win.show()
```

```
# Task 2
win2=QWidget()
layout2=QVBoxLayout()
label=QLabel("")
btn=QPushButton("Click")

def show():
    label.setText("Button Clicked")

btn.clicked.connect(show)
layout2.addWidget(btn)
layout2.addWidget(label)
win2.setLayout(layout2)
win2.show()

# Task 3
win3=QWidget()
layout3=QVBoxLayout()
u=QLineEdit()
p=QLineEdit()
p.setEchoMode(QLineEdit.Password)
layout3.addWidget(u)
layout3.addWidget(p)
layout3.addWidget(QPushButton("Login"))
win3.setLayout(layout3)
win3.show()

sys.exit(app.exec_())
```

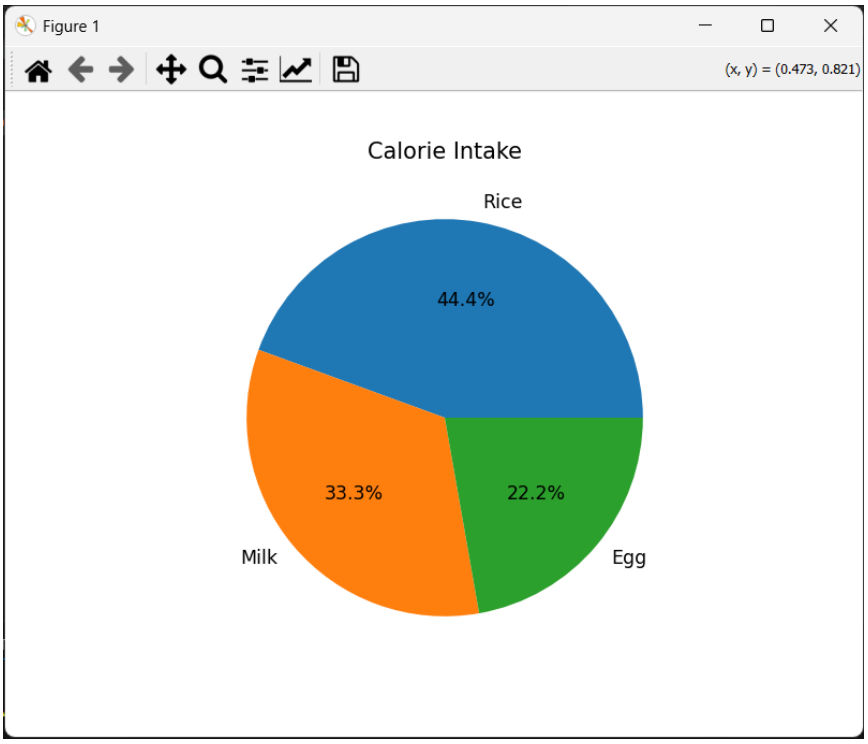
 # NLTK TASKS

```
import nltk
from nltk.tokenize import word_tokenize,sent_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer,WordNetLemmatizer
```

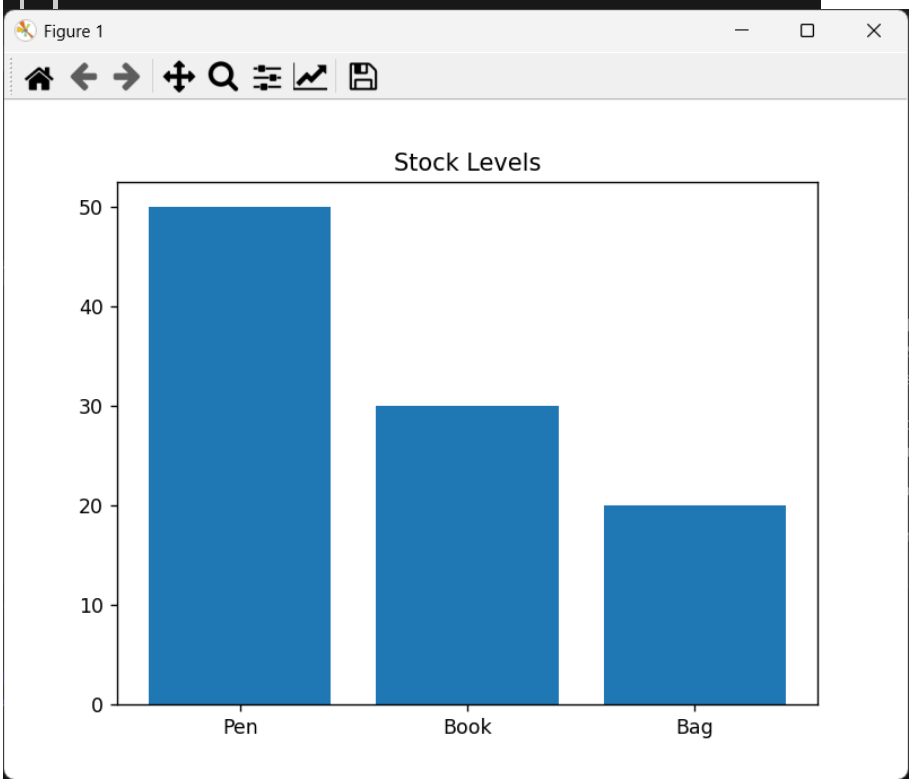
```
# Task 1
text="Natural Language Processing with Python"
print(word_tokenize(text))

# Task 2
para="NLP is interesting. It helps computers understand language."
print(sent_tokenize(para))

# Task 3
words=word_tokenize("This is a simple sentence for stop word removal")
filtered=[w for w in words if w.lower() not in stopwords.words('english')]
print(filtered)

# Task 4
ps=PorterStemmer()
print(ps.stem("running"))

# Task 5
lem=WordNetLemmatizer()
print(lem.lemmatize("running",pos="v"))

# Task 6
tokens=word_tokenize("Python is a powerful programming language")
print(nltk.pos_tag(tokens))
```

Outputs:

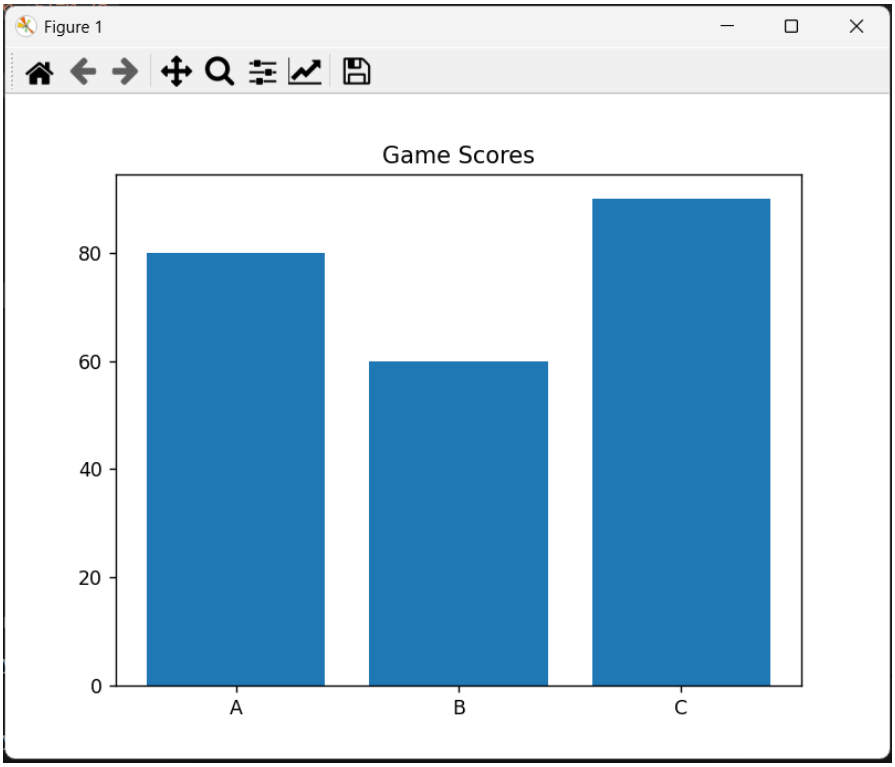
```
Total Calories: 450
```



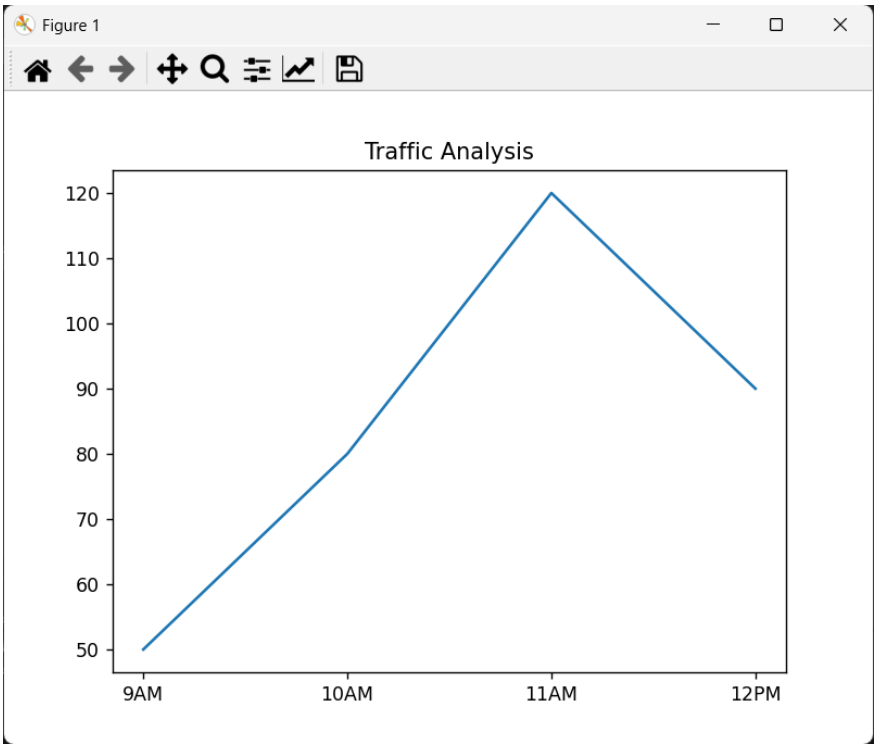

```
Total Stock: 100
```



```
Highest Score: 90
```




```
Peak Time: 11AM
```



```
(.venv) PS D:\Internship\Day33> python num.py
[ 1  2  3  4  5  6  7  8  9 10]
Mean: 30.0
Median: 30.0
Std Dev: 14.142135623730951
[0.06877417 0.24162079 0.03072178 0.56647423 0.18274047 0.2918306
 0.04668905 0.3402678  0.64326475 0.83357807]
[[1 2 3]
 [4 5 6]]
Max: 50
Min: 10
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
[[1. 1. 1.]
 [1. 1. 1.]
 [1. 1. 1.]]
```

```
(.venv) PS D:\Internship\Day33> python pan.py
    Name  Marks
0   Asha     85
1   Ravi     90
2   John     78
3  Meena     88
    Name  Marks Grade
0   Asha     85     A
1   Ravi     90    A+
2   John     78     B
3  Meena     88     A
    Name  Marks Grade
2   John     78     B
0   Asha     85     A
3  Meena     88     A
1   Ravi     90    A+
      Name  Age Department
0    Alice   25         HR
1      Bob   30    Finance
2  Charlie   28         IT
      Name  Age Department
0    Alice   25         HR
1      Bob   30    Finance
2  Charlie   28         IT
      Name  Age Department
0    Alice   25         HR
1      Bob   30    Finance
2  Charlie   28         IT
```

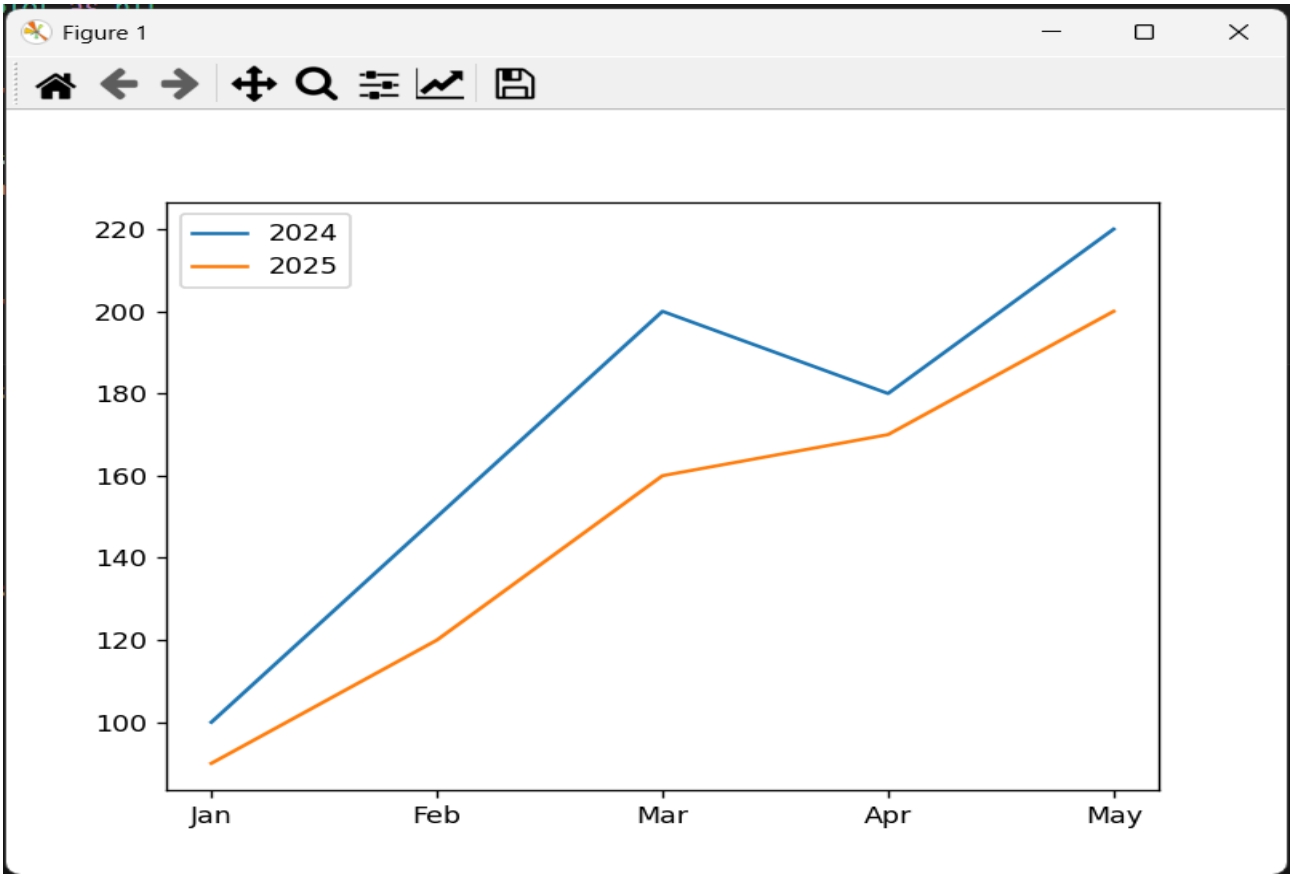


Figure 1
2024
2025
220
200
180
160
140
120
100
Jan
Feb
Mar
Apr
May

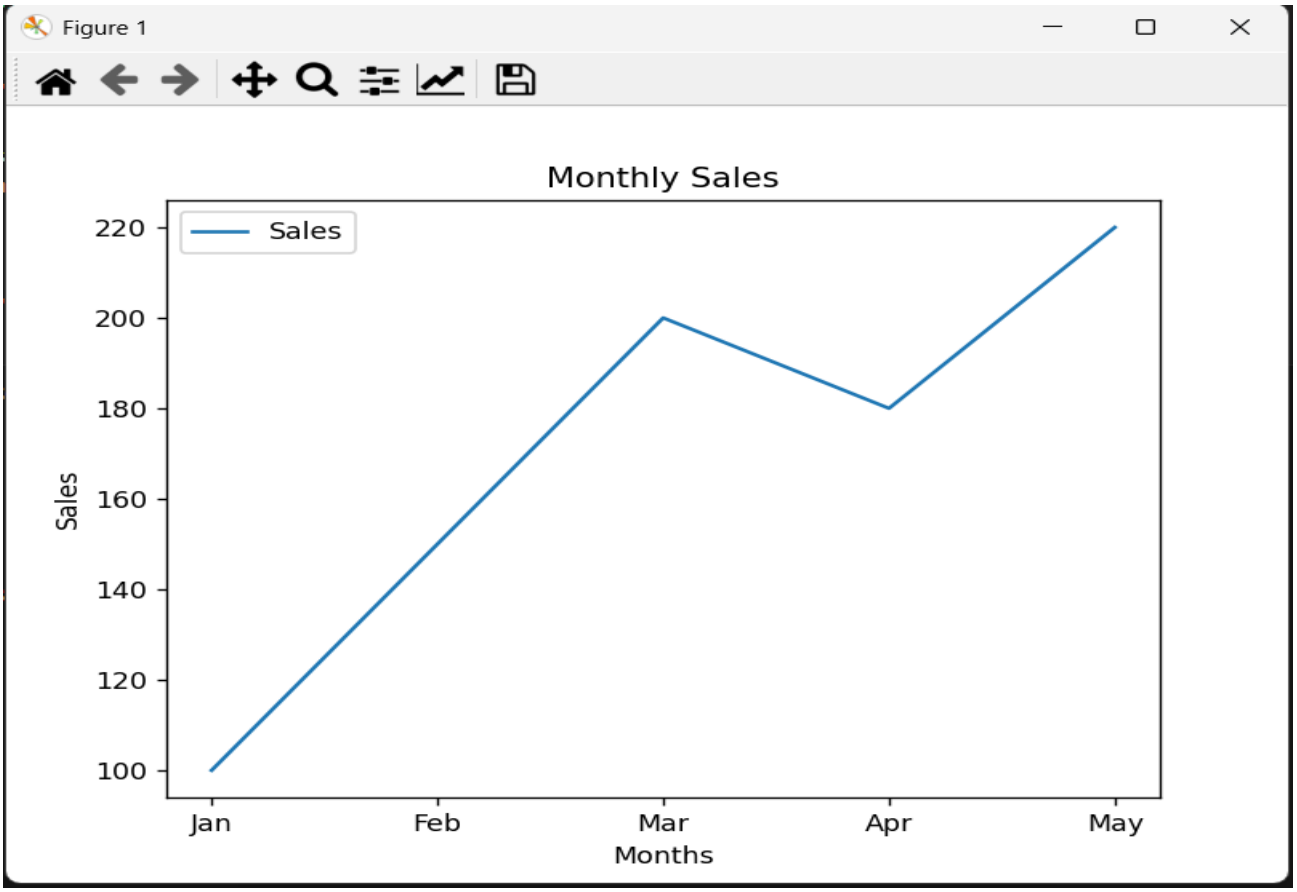


Figure 1
Monthly Sales
Sales
220
200
180
160
140
120
100
Sales
Jan
Feb
Mar
Apr
May
Months

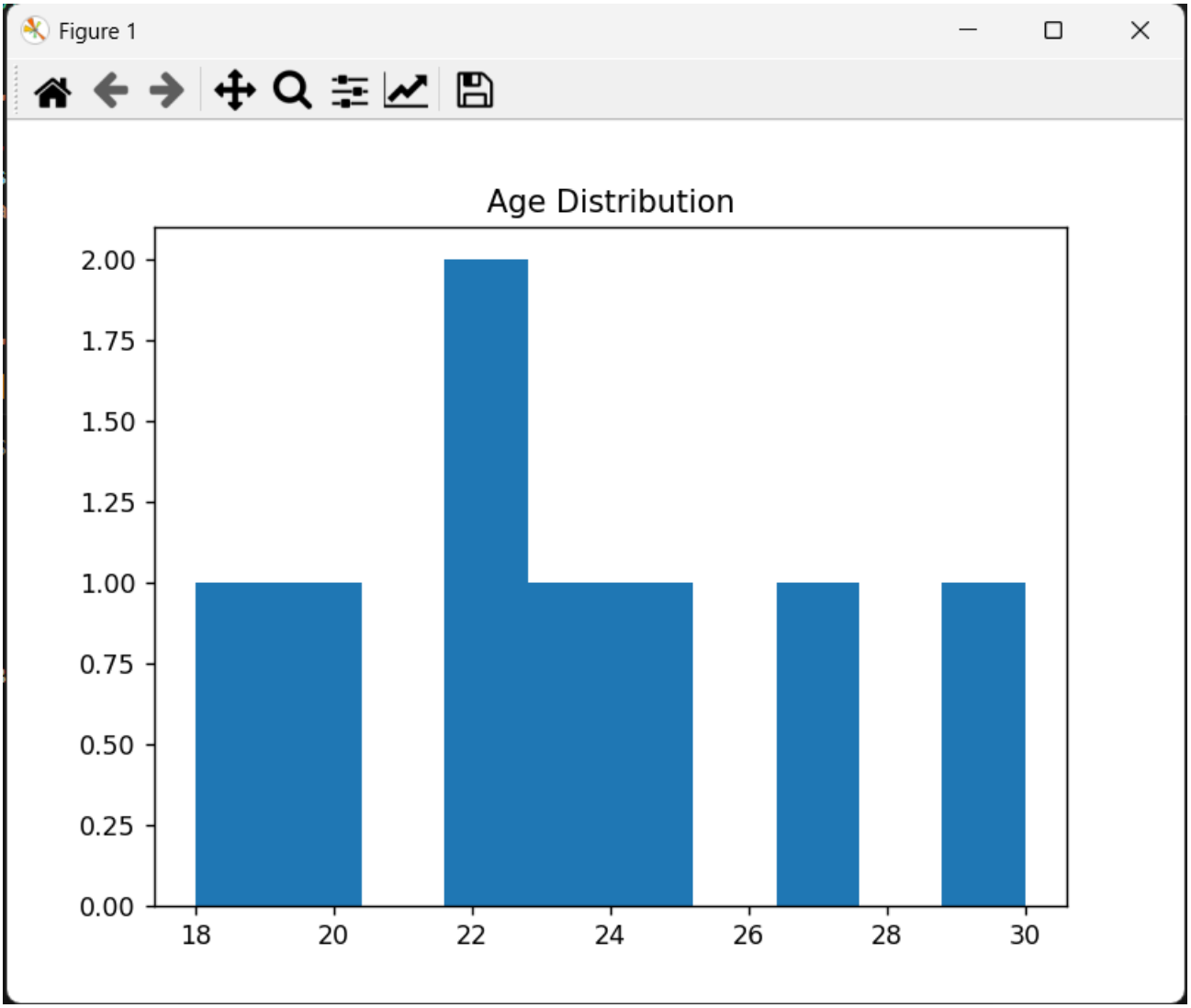


Figure 1
Age Distribution
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
18
20
22
24
26
28
30

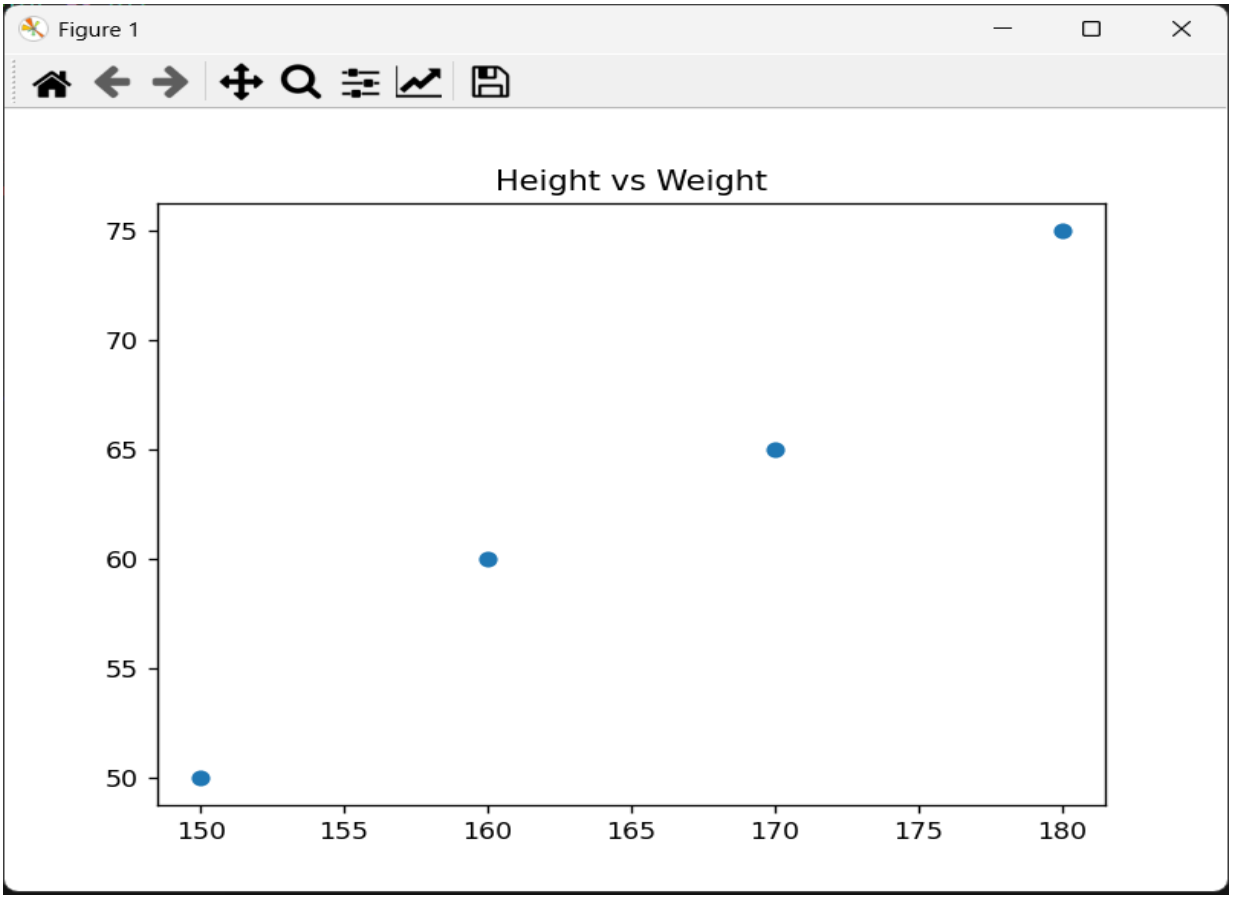


Figure 1
Height vs Weight
75
70
65
60
55
50
150
155
160
165
170
175
180

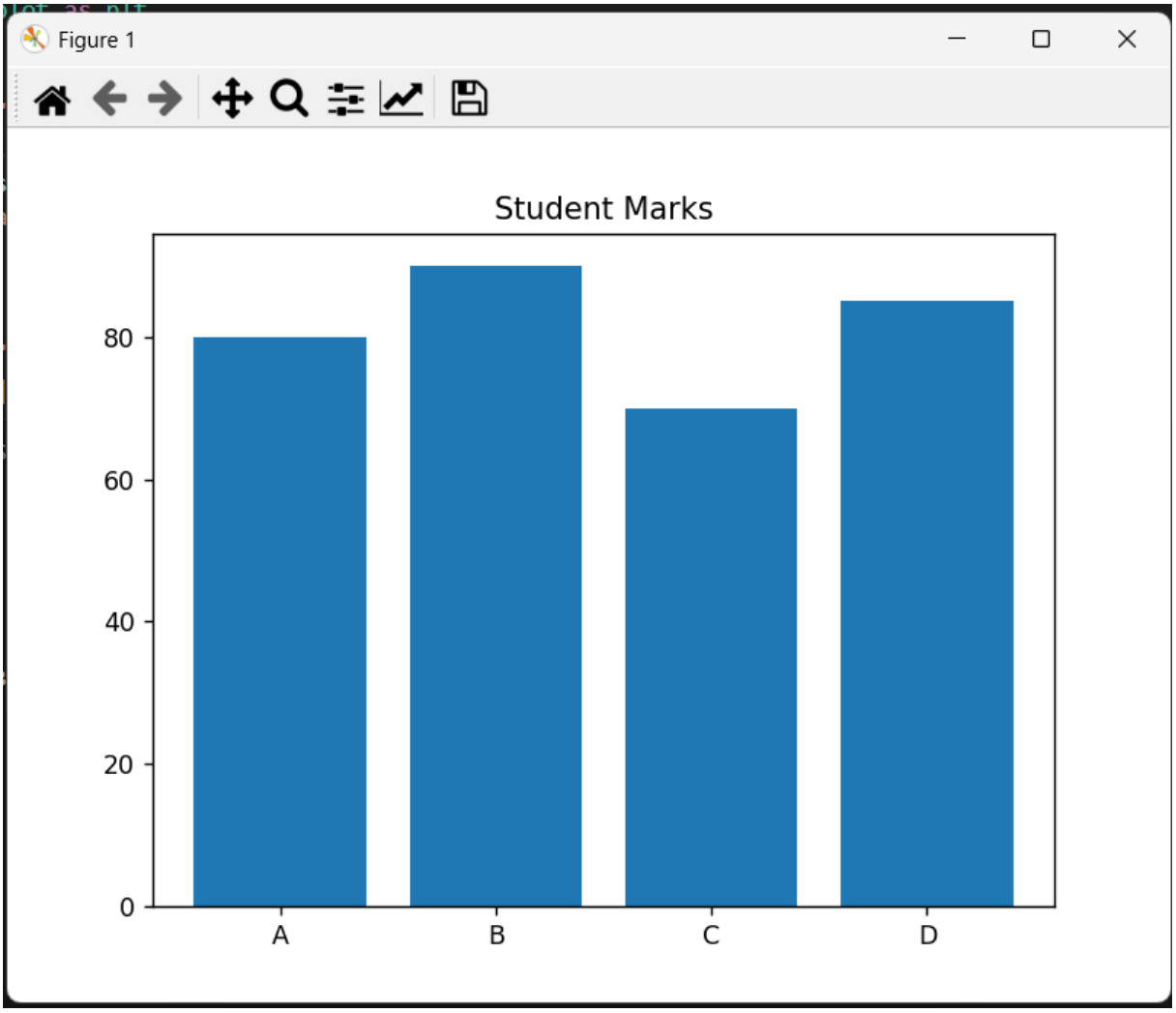


Figure 1
Student Marks
80
60
40
20
0
A
B
C
D

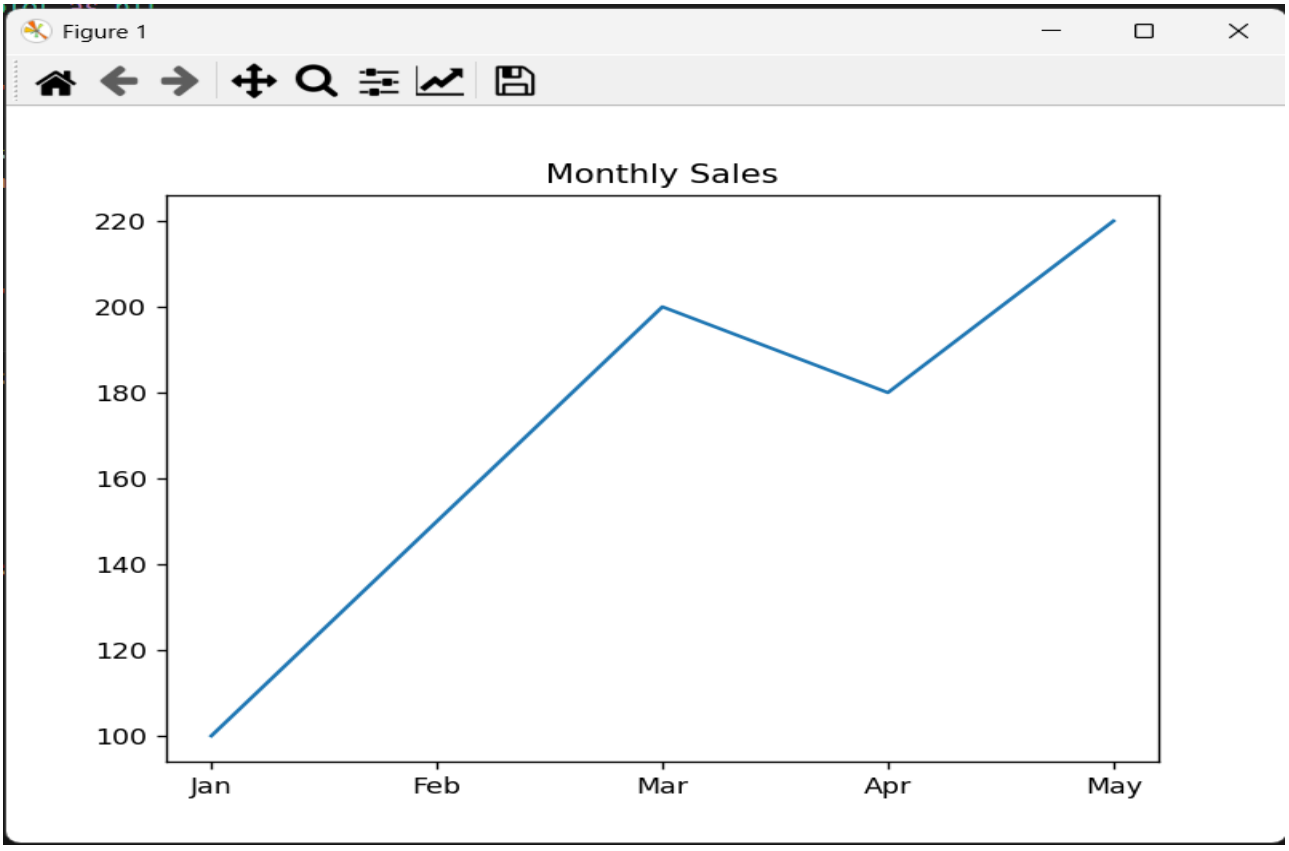


Figure 1
Monthly Sales
220
200
180
160
140
120
100
Jan
Feb
Mar
Apr
May

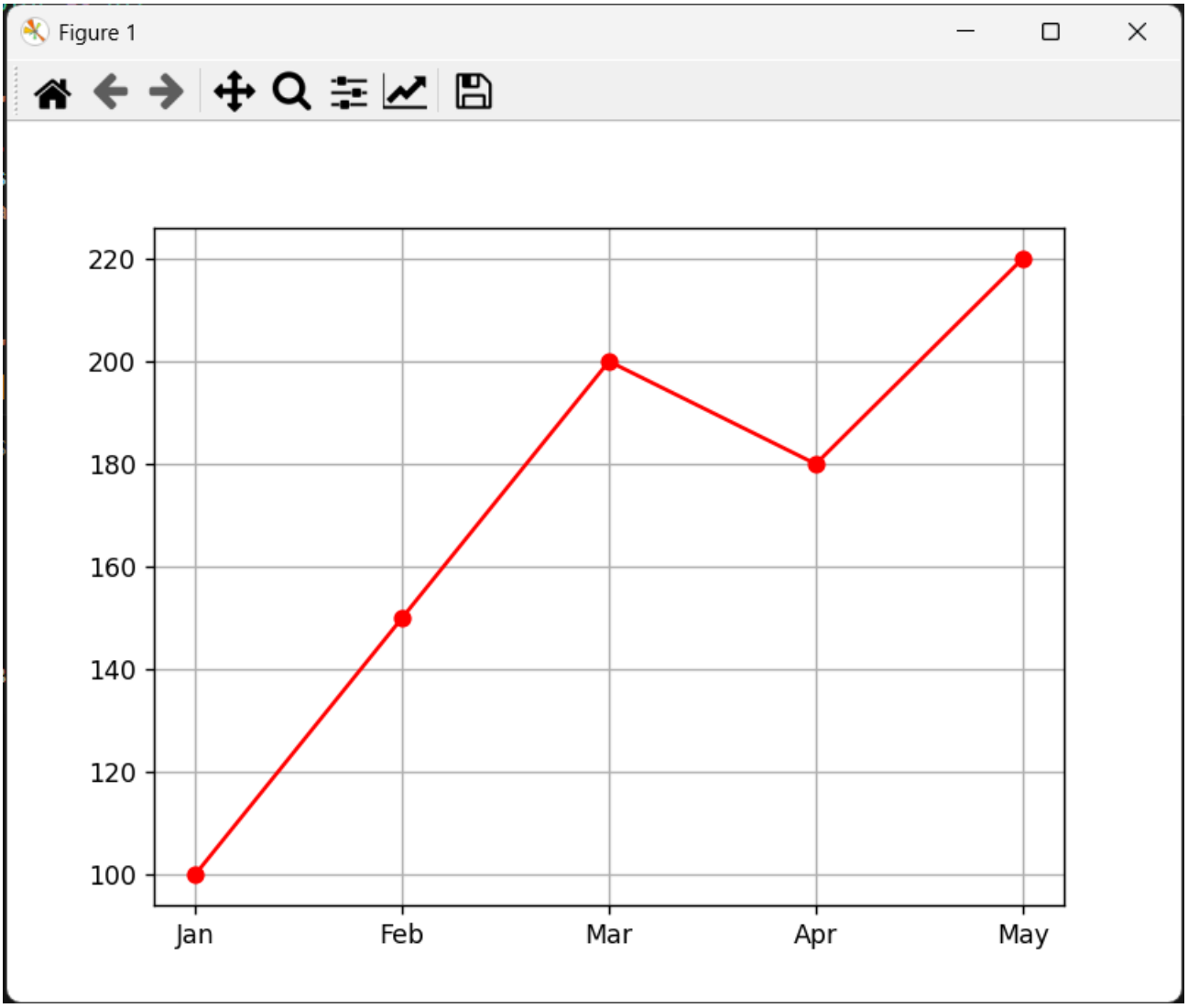
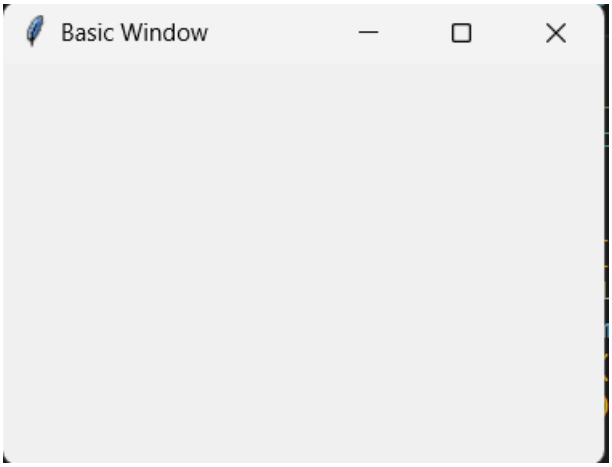
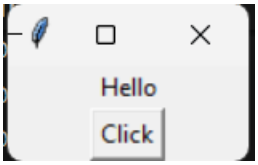
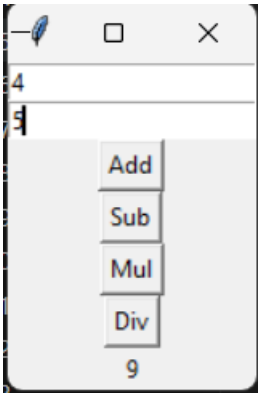
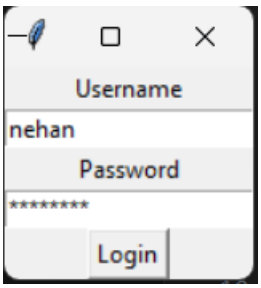


Figure 1
220
200
180
160
140
120
100
Jan
Feb
Mar
Apr
May


Basic Window


Hello
Click

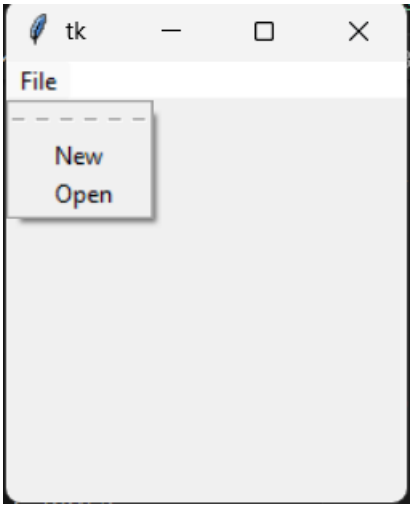

4
5
Add
Sub
Mul
Div
9

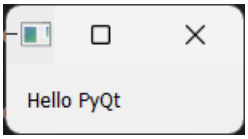

Username
nehan
Password
****
Login

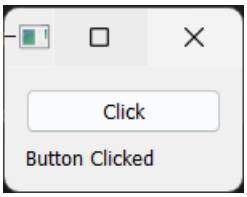
Click
Button Clicked

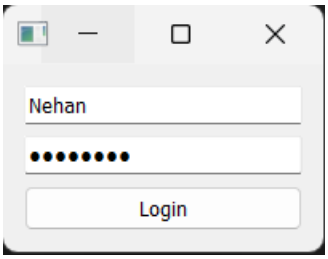
NS

tk
hello world


tk
File
New
Open


Hello PyQt


Click
Button Clicked


Nehan
Login